HIV Analysis

STA198 Team

2026-04-10

```r
rm(list = ls())
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```r
deaths <- read_csv("Data/no_of_deaths_by_country_clean.csv")
```

```
## Rows: 510 Columns: 7
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (3): Country, Count, WHO Region
## dbl (4): Year, Count_median, Count_min, Count_max
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
hiv <- read_csv("Data/no_of_people_living_with_hiv_by_country_clean.csv")
```

```
## Rows: 680 Columns: 7
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (3): Country, Count, WHO Region
## dbl (4): Year, Count_median, Count_min, Count_max
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
art <- read_csv("Data/art_coverage_by_country_clean.csv")
```

```
## Rows: 170 Columns: 11
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (5): Country, Reported number of people receiving ART, Estimated number ...
## dbl (6): Estimated number of people living with HIV_median, Estimated number...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
deaths_clean <- deaths |>
  clean_names() |>
  group_by(country) |>
  summarise(
    deaths = first(count_median),
    .groups = "drop"
  )
```

```r
hiv_clean <- hiv |>
  clean_names() |>
  group_by(country) |>
  summarise(
    hiv_population = first(count_median),
    .groups = "drop"
  )
```

```r
art_clean <- art |>
  clean_names() |>
  select(
    country,
    art_coverage = estimated_art_coverage_among_people_living_with_hiv_percent_median
  )
```

```r
final_df <- deaths_clean |>
  inner_join(hiv_clean, by = "country") |>
  inner_join(art_clean, by = "country")
```

```r
write_csv(final_df, "Data/final_df.csv")
```

```r
glimpse(final_df)
```

```
## Rows: 170
## Columns: 4
## $ country        <chr> "Afghanistan", "Albania", "Algeria", "Angola", "Argenti~
## $ deaths         <dbl> 500, NA, 200, 14000, 1700, 200, 200, NA, NA, 200, NA, 5~
## $ hiv_population <dbl> 7200, NA, 16000, 330000, 140000, 3500, 28000, NA, NA, 6~
## $ art_coverage   <dbl> 13, NA, 81, 27, 61, 53, 83, NA, NA, 52, NA, 22, 50, 59,~
```

Description

I created a merged, final dataset to use for analysis with cleaned and standardized columns (contains 170 countries and 4 variables). Each row represents a single country. The dataset was constructed by merging three country-level datasets sourced from WHO-based HIV/AIDS indicators available on Kaggle. All variables are continuous numerical estimates, and represent median values reported in the original datasets. The variables included in the final dataset are: - HIV/AIDS deaths (deaths) - Number of people living with HIV

(hiv_population) - ART coverage among people living with HIV (art_coverage)

```r
colSums(is.na(final_df))
```

```
##        country         deaths hiv_population   art_coverage
##              0             37             32             34
```

```r
colMeans(is.na(final_df))
```

```
##        country         deaths hiv_population   art_coverage
##      0.0000000      0.2176471      0.1882353      0.2000000
```

Description

The dataset contains missing values across all three main variables: HIV/AIDS deaths (37 missing, ~21.8%), number of people living with HIV (32 missing, ~18.8%), and ART coverage (34 missing, 20%); no missing data for countries. This is most likely due to incomplete reporting in certain countries within the original WHO-based dataset (real-world data is often incomplete). Missing data was not cleaned or removed, just be wary of it and decide what to do with it in the next stages (if anything).

```r
final_df |>
  arrange(desc(deaths)) |>
  slice(1:10)
```

```
## # A tibble: 10 x 4
##    country                     deaths hiv_population art_coverage
##    <chr>                        <dbl>          <dbl>        <dbl>
##  1 South Africa                 71000        7700000           62
##  2 Mozambique                   54000        2200000           56
##  3 Nigeria                      53000        1900000           53
##  4 Indonesia                    38000         640000           17
##  5 Kenya                        25000        1600000           68
##  6 United Republic of Tanzania  24000        1600000           71
##  7 Uganda                       23000        1400000           72
##  8 Zimbabwe                     22000        1300000           88
##  9 Cameroon                     18000         540000           52
## 10 Thailand                     18000         480000           75
```

Description

Shows top 10 countries with the most HIV/AIDS deaths.

```r
final_df |>
  filter(!is.na(deaths)) |>
  arrange(deaths) |>
  slice(1:10)
```

```
## # A tibble: 10 x 4
##    country                deaths hiv_population art_coverage
##    <chr>                   <dbl>          <dbl>        <dbl>
##  1 Barbados                  100           3000           50
##  2 Bhutan                    100           1300           37
##  3 Bosnia and Herzegovina    100            500           67
##  4 Bulgaria                  100           3500           41
##  5 Cabo Verde                100           2400           89
##  6 Comoros                   100            200           79
##  7 Croatia                   100           1600           75
```

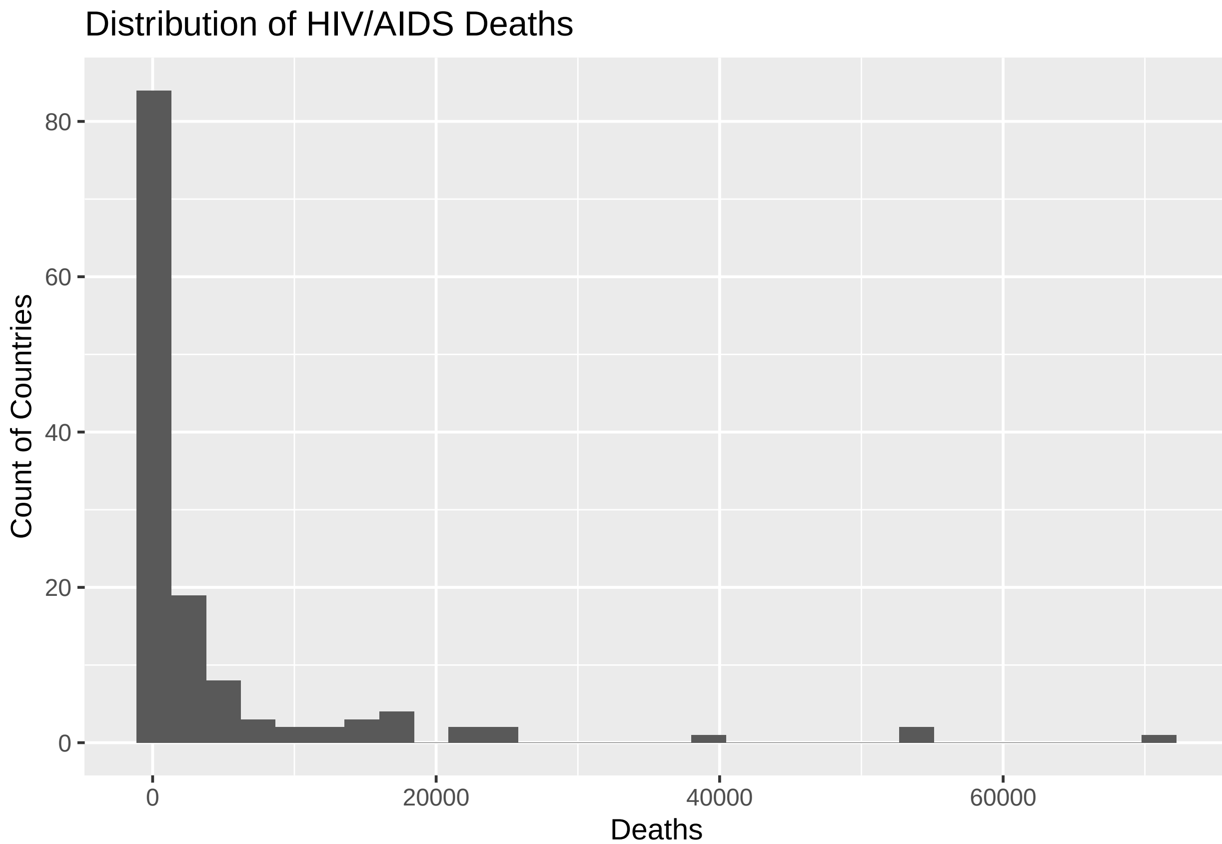
```
##   8 Czechia                      100           4400           60
##   9 Denmark                      100           6200           89
## 10 Estonia                      100           7400           59
```

Description

Shows top 10 countries with the least HIV/AIDS deaths.

```
ggplot(final_df, aes(x = deaths)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of HIV/AIDS Deaths", x = "Deaths", y = "Count of Countries")
```

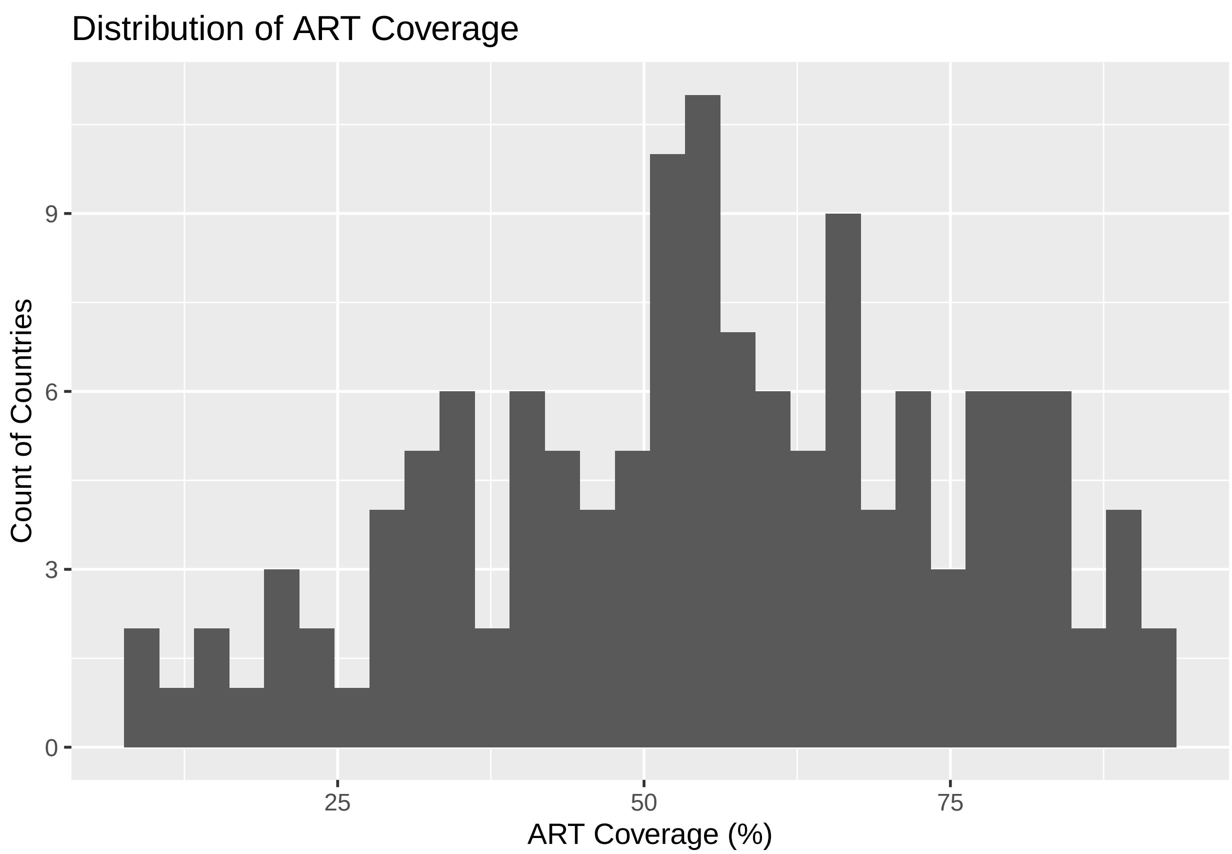
```
## Warning: Removed 37 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



Description Shows the distribution of deaths across countries.

```
ggplot(final_df, aes(x = art_coverage)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of ART Coverage", x = "ART Coverage (%)", y = "Count of Countries")
```

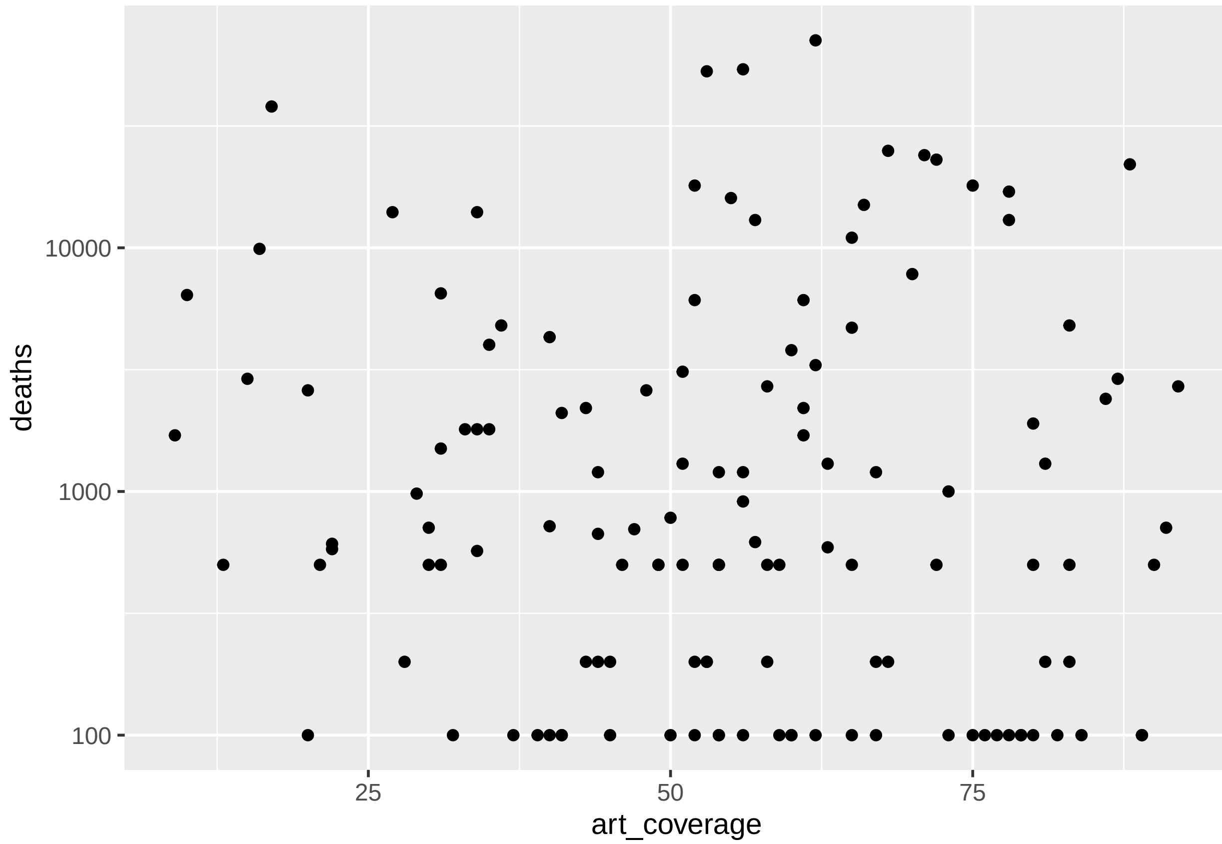
```
## Warning: Removed 34 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



Description Shows the distribution of treatment access (ART coverage %) across countries.

```
ggplot(final_df, aes(x = art_coverage, y = deaths)) +
  geom_point() +
  scale_y_log10()
```

```
## Warning: Removed 39 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Description Shows the relationship between ART coverage (%) and HIV/AIDS deaths across countries, where each point represents a country. Note that I used log scaling for the y-axis to account for scaling issues and compression of the graph.